Family ecological influences on early childhood development in Shanghai, China: a cross-sectional study

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Resumo (Abstract)

Background Studies regarding the prevalence of early childhood development (ECD) delays and comprehensive evidence of family ecological influences in urban regions of China are scarce. We aimed to include multilevel factors in Bronfenbrenner's family ecosystem model and examine the effect of each factor on ECD among 1–36-month-old children. Methods A questionnaire was designed to collect demographic characteristics and family ecological factors from the microsystem and exosystem. The questionnaires were distributed online to caregivers in Shanghai from February 6, 2022, to March 29, 2024. Results A total of 557 questionnaires were distributed, and 534 valid responses were obtained. In total, 16.7% of children confirmed developmental delay in at least one developmental domain. Boys, children born prematurely with oxygen supplementation, those with fewer child books, parents with lower marital happiness, less frequent parent–child activities, mothers' screen use and associated neglect, and poorer family functioning were risk factors for early developmental delays, particularly affecting language, problem-solving, personal-social, and social-emotional domains. Conclusion Children aged 1–36 months in Shanghai face a severe risk of ECD delays. The identified family ecological factors, as conceptualized within the family ecosystem theory and the Nurturing Care Framework, are measurable and may represent potential targets for early identification and intervention. Supplementary Information The online version contains supplementary material available at https://doi.org/10.1186/s12887-026-07167-w .

Background

Early childhood development (ECD) encompasses cognitive, physical, language, temperament, socio-emotional, and motor development from conception to eight years of age [1 , 2], and is fundamental to a child's right to survive and thrive [3]. Thi s study focuses on children aged 0–36 months. The period from pregnancy to 3 years of age is particularly critical, as approximately 80% of brain development occurs during this time [3]. ECD is strongly associated with lifelong health, educational attainment, and economic productivity, thereby influencing intergenerational cycles of health and social well-being [4 , 5].

ECD is shaped by multidimensional influences operating across biological, familial, social, and environmental contexts. Previous research has primarily examined individual determinants of ECD, including demographic characteristics [6], birth-related factors [7], nutrition [8], responsive caregiving [9], early learning opportunities [10], family environment [11], family relationships [12], socioeconomic status [13], intergenerational influences [14], life-course changes [15], and maternal mental health [16]. However, these studies have largely evaluated single factors in isolation, despite the frequent co-occurrence and interaction of multiple risks [17]. This fragmented approach may yield biased estimates and limit effectiveness and sustainability of interventions. For example, the protective effect of a high-quality parent–child relationship may be attenuated in the context of poor marital functioning [18].

Because ECD unfolds within dynamic social, cultural, and temporal environments [19 , 20], comprehensive investigation across multiple contextual levels is essential [21]. Key determinants identified in prior studies can be conceptualized within five domains: nurturing care, family relationships, socioeconomic characteristics and intergenerational culture. These domains align closely with Bronfenbrenner's ecological systems theory [22 , 23].

Bronfenbrenner's theory [24 – 26] posits that child development is shaped by interacting environmental systems, including the microsystem, mesosystem, exosystem, macrosystem, and chronosystem. Due to data accessibility constraints, this study focused on selected factors within the microsystem and exosystem. The microsystem comprises the child's immediate environment and relationships, including caregiving practices and family functioning. Within this system, nurturing care reflects parenting behaviors, whereas family functioning reflects relational quality and adaptability. The Nurturing Care Framework, proposed by the World Health Organization (WHO) and United Nations Children's Fund (UNICEF) [3 , 27], emphasizes five core components: health, nutrition, responsive caregiving, early learning opportunities, and safety and security. Family relationships encompass interactions among family members, including parent–child and spousal relationships, which evolve over time and may involve conflict [28]. The manner in which families manage conflict and adapt to developmental changes reflects overall family functioning. The McMaster model conceptualizes family functioning across seven domains: problem solving, communication, roles, affective responsiveness, affective involvement, behavioral control, and general functioning [29]. Meanwhile, in contemporary parenting environments, screen exposure has become increasingly significant, exerting a dual influence on ECD [30]. The exosystem includes broader contextual factors that indirectly influence development [30], such as parental employment and socioeconomic conditions.

Most existing research has focused on microsystem factors and single ECD domains, particularly nurturing care and family relationships. Preterm birth, for example, has been associated with poorer language [31 , 32] and cognitive [33] outcomes. Frequent parent–child interaction has been linked to improved cognitive, motor, socio-emotional, and language development [34 – 36]. Limited access to books and play materials has been associated with adverse developmental outcomes [10 , 37 , 38]. Despite these findings, few studies have examined family functioning within the ecological framework [39]. Research has also explored chronosystem influences, such as parental divorce, on ECD [40]. However, comparatively little attention has been given to exosystem and macrosystem factors. Family poverty [41 , 42] and multigenerational caregiving, for example, have demonstrated context-dependent associations with ECD outcomes [14]. Importantly, most previous studies have only examined a limited set of factors within a single subsystem, lacking a comprehensive analysis of factors across different subsystems [18]. Furthermore, most evidence derives from high-income countries [35 , 43 – 45] or rural regions of middle-income countries, including rural China [10 , 14 , 37 , 39 , 42 , 46 – 49]. Given socioeconomic disparities [42 , 46] and urban–rural differences [44] in ECD outcomes, these findings may not generalize to highly developed urban settings such as Shanghai [50 , 51].

Additionally, prior studies have often emphasized specific developmental domains rather than comprehensively evaluating all dimensions of ECD. In view of this limitation, the present study employed the Ages & Stages Questionnaires to assess child development comprehensively across multiple domains, including cognitive, language, motor, social-emotional and mental health development. To address this gap, this study focused on microsystem factors and selected factors from the exosystem, examining their associations with multiple ECD domains among children aged 1–36 months in Shanghai, China.

Methods

Study design

This study aimed to explore the influence of family ecosystem factors on ECD in Shanghai. The included variables comprised nurturing care, family relationships, family functioning, home literacy environment and screen exposure from the microsystem, as well as family socioeconomic status from the exosystem. Other factors from the mesosystem, exosystem (e.g., access to childcare and community services), and macrosystem (e.g., culture and values) were not included, as pilot survey participants reported little attention to and difficulty answering these aspects. Moreover, the cross-sectional design precluded assessment of chronosystem influences. The corresponding ecological system levels, factors, and their explanation are summarized in Table 1 .

Table 1 Classification and interpretation of study variables according to Bronfenbrenner's ecological systems theory Ecological system levels Factors Explanation Demographic characteristics Sex Child's sex Age Child's age in months Preterm birth Preterm birth occurs before 37 weeks of gestation Preterm birth with oxygen supplementation at birth Preterm birth with low birth weight Intergenerational support Grandparental involvement in childcare Microsystem Nurturing care Breastfeeding Never been breastfed / Continuous breastfeeding for at least 6 months Secure and safe level of family parenting environment Assessed by the 0–6 Years Child Home Nurture Environment Questionnaire (CHNEQ) Dialect use Use of dialect or Mandarin by parents when speaking to the child Parent-child shared activities Weekly frequency of parent-child shared activities Screen exposure and parental technoference Screen exposure Child's weekly frequency of screen exposure Child's average daily screen exposure time in the past week Maternal screen neglect Mother uses digital devices while caring for or conversing with child Mother tends to ignore child while using digital devices Child attempts to attract mother's attention during her device use Early screen initiation Child's first exposure to screens within 6 months Home Literacy Environment Number of adult books at home Number of children's books at home Mother's frequency of reading newspapers or books Number of sound-producing toys at home Initiated reading or storytelling to child within 6 months Family relationships Siblings Having siblings marital arguments Frequency of marital arguments in the past 3 months marital happiness Self-rated marital happiness Family functioning Assessed by family APGAR score Exosystem Socioeconomic status Maternal / Paternal educational level Maternal / Paternal employment status Household monthly income (CNY)

Family ecological factors

A questionnaire was designed to collect the basic demographic characteristics and family ecosystem factors of 1–36-month-old children and their families. Basic demographic characteristics included the child's sex, monthly age, preterm birth, preterm birth with oxygen supplementation at birth, low birth weight [7 , 32 , 33], and intergenerational support [14 , 52 , 53]. Microsystem factors included nurturing care, family relationships, family functioning, home literacy environment and screen exposure. Nurturing care factors were measured using the 0–6 Years Child Home Nurture Environment Questionnaire (CHNEQ) developed by He Shousen (Cronbach's $\alpha = 0.918$) [54]. Nurturing care factors included adequate nutrition [8 , 55], responsive caregiving [56], opportunities for early learning [10 , 37 , 38 , 49 , 57 , 58], and security and safety. Family relationships were assessed using the Dyadic Adjustment Scale developed by Spanier, which has demonstrated extensive reliability and validity in the Chinese context (Cronbach's $\alpha = 0.85$) [59]. Family relationships focused on nuclear families, including sibling presence [47 , 60 , 61], parent–child shared activity frequency [34 – 36 , 45], marital argument frequency, and marital happiness [59 , 62

, 63]. Family functioning was evaluated using the Family APGAR Index (APGAR) (Cronbach's α = 0.86) [64], calculated as the sum of the five item scores, with lower scores indicating worse family functioning. Family reading environment was assessed using indicators derived from the Home Literacy Environment Questionnaire (HLEQ) developed by Umek et al. (Cronbach's α = 0.90) [65]. Screen exposure-related behaviors were measured using the Parent version of the parent–child technoference scale revised by Stockdale et al. (Cronbach's α = 0.87) [56]. Exosystem factors included maternal and paternal educational level [66 – 68], maternal and paternal employment status [69], and household monthly income [41]. The questionnaire had acceptable reliability and validity.

ECD outcome assessment

The insidious nature of ECD delays makes it easy for caregivers to overlook the optimal intervention period, while constraints in accessing professional evaluation resources frequently result in a "wait-and-see" attitude, thereby substantially limiting early detection, diagnosis, and effective interventions to improve ECD outcomes [15 , 70 , 71]. Therefore, a simple population-level measurement of caregivers might provide more opportunities for better ECD outcomes. The Ages & Stages Questionnaire–3rd edition (ASQ-3) is a cost-effective tool [72 – 74] that enables caregivers to self-assess their children's ECD status. It is an accurate and valid tool for caregivers to conduct a preliminary screening of ECD delays or to determine if a referral for further diagnosis is necessary [75 – 78]; the Chinese version has also been validated [79]. We utilized the ASQ-3 to evaluate the ECD levels in five dimensions: communication, gross motor skills, fine motor skills, problem-solving skills, and personal-social skills. A total score lower than two standard deviations (SDs) from the mean (lower than the cutoff) indicates high probabilities of developmental delay; 1–2 SDs, close to cutoff; and higher than the mean SD (above the cutoff), normal development.

Additionally, considering the growing importance of early social–emotional issues [80], we employed the Ages and Stages Questionnaires: Social-Emotional, Second Edition (ASQ: SE-2) to specifically assess these problems. ASQ: SE-2 is a tool completed by the parent that measures social–emotional problems [81]. This instrument includes seven behavioral dimensions: self-regulation, compliance, adaptive functioning, autonomy, affect, social communication, and interaction. A total score lower than the cutoff (< P75) indicates normal development; close to the cutoff (≥ P75), substantial concerns about social–emotional development; and higher than the cutoff (≥ P90), high probabilities of development delay.

Data collection

The required sample size was at least 186 responses, which was calculated based on a confidence level of 95%, permissible error of 0.05, and a non-achievement rate for ECD milestones of 14% in Shanghai, China [82]. After routine check-ups at community health service centers, community pediatricians introduced the study purpose to eligible families and obtained informed consent. Enrolled parents filled out the questionnaires via an online link with standardized instructions, and could consult the community pediatricians online for any questions they encountered. The recruitment process is summarized in Fig. 1 .

Fig. 1 Flowchart of participant recruitment

The family ecological factors questionnaires ASQ-3 and ASQ: SE-2 were distributed online from February 6, 2022, to March 29, 2024. Convenience sampling was also performed. We included children aged 1–36 months and primary maternal caregivers willing to participate in the study and

who signed an informed consent form. We excluded children with congenital, chronic, or major diseases; caregivers unable to complete the self-assessment of their ECD status or complete the online questionnaires; and children living in Shanghai for < 10 months within a year. All questionnaires were completed independently by the children's mothers. Ultimately, 557 questionnaires were distributed, and 534 valid responses were obtained (response rate: 95.9%).

Representativeness of the study sample

Compared with the general population of Baoshan District, our study sample had a higher proportion of mothers with a bachelor's degree or above and a higher proportion of households with a monthly income above 20,000 CNY, suggesting that our convenience sample may overrepresent socioeconomically advantaged families. The data were derived from the Shanghai Statistical Yearbook and the Baoshan District Statistical Yearbook (2023), as detailed in Table 2 .

Table 2 Source population vs. study sample: Demographic and socioeconomic characteristics Category Subgroup Shanghai (2.485 million) Baoshan District (0.227 million) Study Sample (N = 534) Mean age at childbirth 32.6 31.2 31.0 Sex of child Male 51.6% 50.3% 52.2% Female 48.4% 49.7% 47.8% Educational level High school or below 40.3% 45.2% 18.0% Associate degree 25.2% 26.3% 31.1% Bachelor's degree 27.9% 24.1% 46.1% Master's degree 5.8% 4.5% 4.5% Doctorate 0.9% 0.1% 0.4% Household monthly income (CNY) ≤ 5000 12.0% 14.2% 4.1% 5001–10,000 28.3% 30.2% 18.9% 10,001–20,000 35.2% 33.8% 48.4% > 20,000 24.6% 21.9% 28.5% Age group distribution of children (0–3 years) 0–6 months 16.9% 16.6% 3.2% 7–12 months 16.5% 16.7% 37.5% 13–18 months 16.8% 16.6% 22.9% 19–24 months 16.9% 16.2% 18.0% 25–30 months 16.9% 16.7% 16.7% 31–36 months 15.9% 17.2% 1.7%

Statistical analysis

Categorical variables are described as numbers with percentages, and continuous variables are described as mean ± SD. ECD outcomes were analyzed as categorical variables. Statistical analyses were performed using IBM SPSS Statistics version 26.0 (IBM Corp., Armonk, NY). All demographic and family ecological factors were compared among different ECD statuses using the Spearman test for one continuous variable and one ordinal variable, and the Cochran–Mantel–Haenszel (CMH) statistic for binary, categorical, and ordinal variables in R×C tables. The above analyses were performed using SAS version 8.2 (SAS Institute Inc., Cary, NC). Provided that the assumptions of parallel lines and the absence of severe multicollinearity were met, ordinal logistic regression analysis was conducted to explore family ecological influences on ECD. Otherwise, a multinomial logistic regression model was used. These analyses were performed using the R version 4.2.3 (R Foundation for Statistical Computing, Vienna, Austria). Candidate variables for logistic regression were screened by univariate analysis ($P < 0.1$), followed by selection of key variables based on professional judgment and literature. To avoid overfitting, we applied the EPV principle with an acceptable threshold of EPV ≥ 5. Variables that failed univariate screening but were considered important were only reported in univariate analyses and not forcibly included in multivariate models. The EPV for the six models were 8, 20, 10, 6, 10, and 10, all ≥ 5. Stepwise selection was not used, and collinearity diagnostics were performed. All tests were two-tailed, and statistical significance was set at $p < 0.05$.

Ethics statement

This study was approved by the Ethics Committee of the School of Public Health, Shanghai Jiao Tong University School of Medicine (SJUPN-202109). In accordance with Chinese laws and

regulations, informed consent has been obtained from the legal guardians of the children participating in the study.

Results

Basic demographic and family ecological characteristics

Among the 534 children (average age, 13.8 ± 6.9 months), there were more boys. Nearly 5% of the children were born preterm, approximately 1% of the children were born preterm and received oxygen supplementation at birth, and approximately 4% of the children were born preterm with low birth weight. Nearly 10% of the children were never breastfed, and over 70% were continuously breastfed for at least 6 months. Over 20% of mothers ignored their children when using digital devices, over 60% attempted to attract their mothers' attention during device use, and over 35% used digital devices when caring for their children. The local dialect was used by 10% of parents when speaking to their children. Most families had 1–2 sound-producing toys, 1–10 adult books, and 1–10 children's books. Approximately half of the mothers had never read newspapers or books. Approximately 20% of the children were exposed to screens every day, and approximately 5% had a daily screen time of > 1 h. Over 70% of parents thought that the safety and security of the family parenting environment was fair. Approximately 40% of children have siblings. Approximately half of the parents engaged in parent–child shared activities 3–4 times per week. Most parents rarely had arguments over the previous 3 months and reported a relatively high level of marital happiness. Nearly half of parents had a bachelor's degree. Approximately 75% of mothers were employed, and nearly 99% of fathers were employed. Most families had a monthly household income of > 20,000 CNY. Nearly 80% of the families received caregiving support from grandparents. Approximately 40% of children initiated reading or storytelling within 6 months, and 20% of children were first exposed to screens within 6 months (Table 3).

Table 3 Basic demographic and family ecological characteristics Characteristic N / \documentclass[12pt]{minimal} \usepackage{amsmath} \usepackage{wasysym} \usepackage{amsfonts} \usepackage{amssymb} \usepackage{amsbsy} \usepackage{mathrsfs} \usepackage{upgreek} \setlength{\oddsidemargin}{-69pt} \begin{document}$$\:\stackrel{-}{X}$$\end{document} %/SD Sex Male 279 52.2 Female 255 47.8 Age (month) 13.8 6.9 Preterm birth Yes 25 4.7 No 509 95.3 Preterm birth with oxygen supplementation at birth Yes 6 1.1 No 528 98.9 Preterm birth with low birth weight Yes 23 4.3 No 511 95.7 Never been breastfed Yes 52 9.7 No 482 90.3 Continuous breastfeeding for at least 6 months Yes 392 73.4 No 142 26.6 Mother tends to ignore child while using digital devices Strongly disagree 166 31.1 Disagree 153 28.7 Uncertain 88 16.5 Agree 91 17.0 Strongly agree 36 6.7 Child attempts to attract mother's attention during her device use Strongly disagree 40 7.5 Disagree 50 9.4 Uncertain 107 20.0 Agree 232 43.4 Strongly agree 105 19.7 Mother use digital devices while caring for or conversing with child Strongly disagree 102 19.1 Disagree 130 24.3 Uncertain 109 20.4 Agree 152 28.5 Strongly agree 41 7.7 Language used by mother when speaking to child Mandarin 477 89.3 Local dialect 57 10.7 Language used by father when speaking to child Mandarin 466 87.3 Local dialect 68 12.7 Number of sound-producing toys at home None 24 4.5 1–2 204 38.2 3–4 203 38.0 5–6 58 10.9 ≥ 7 45 8.4 Number of adult books at home None 24 4.5 1–10 215 40.3 11–20 119 22.3 21–40 78 14.6 41–60 33 6.2 > 60 65 12.2 Number of children's books at home None 31 5.8 1–10 198 37.1 11–20 108 20.2 21–40 74 13.9 41–60 43 8.1 > 60 80 15.0 Mother's frequency of reading newspapers or books Almost never 260 48.7 1–2 times / week 217 40.6 3–4 times / week 25 4.7 5–6 times / week 3 0.6 Daily 29 5.4 Child's weekly frequency of screen exposure Almost never 178 33.3

1–2 times / week 134 25.1 3–4 times / week 72 13.5 5–6 times / week 41 7.7 Daily 109 20.4 Child’s average daily screen exposure time in the past week None 177 33.1 < 30 min 263 49.3 30 min–1 h 66 12.4 1–2 h 19 3.6 > 2 h 9 1.7 Secure and safe level of family parenting environment Poor 4 0.7 Fair 383 71.7 Good 146 27.3 Excellent 1 0.2 Having siblings Yes 192 36.0 No 342 64.0 Frequency of parent-child shared activities Never 2 0.4 Rarely (1–2 times / week) 53 9.9 Sometimes (3–4 times / week) 262 49.1 Often (5–6 times / week) 180 33.7 Daily 37 6.9 Frequency of marital arguments in the past 3 months Never 154 28.8 Rarely 299 56.0 Sometimes 77 14.4 Often 4 0.7 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 1 2 0.4 2 4 0.7 3 4 0.7 4 44 8.2 5 143 26.8 6 211 39.5 7 126 23.6 Family functioning (APGAR) score 7.5 2.6 Maternal educational level High school or below 98 18.4 Associate degree 160 30.0 Bachelor’s degree 252 47.2 Master’s degree 23 4.3 Doctorate 1 0.2 Paternal educational level High school or below 94 17.6 Associate degree 172 32.2 Bachelor’s degree 240 44.9 Master’s degree 25 4.7 Doctorate 3 0.6 Maternal employment status Employed 404 75.7 Unemployed 130 24.3 Paternal employment status Employed 526 98.5 Unemployed 8 1.5 Household monthly income (CNY) ≤ 5000 22 4.1 5001–10,000 101 18.9 10,001–15,000 133 24.9 15,001–20,000 126 23.6 > 20,000 152 28.5 Grandparental involvement in childcare Yes 424 79.4 No 110 20.6 Initiated reading or storytelling to child within 6 months Yes 216 40.4 No 318 59.6 Child’s first exposure to screens within 6 months Yes 107 20.0 No 427 80.0 Development delay in one domain of ECD 89 16.7 Suspected development delay in one domain of ECD 198 37.1 Development delay in two domains of ECD 17 3.2 Suspected development delay in two domains of ECD 70 13.1 Development delay in three domains of ECD 4 0.7 Suspected development delay in three domains of ECD 22 4.1 Development delay in four domains of ECD 2 0.4 Suspected development delay in four domains of ECD 4 0.7 Development delay in five domains of ECD 1 0.2 Suspected development delay in five domains of ECD 1 0.2

ECD levels and differences among children

In total, 16.7% of children had confirmed developmental delay in at least one domain, and 37.1% had suspected delay (Table 3). Additionally, 5.1%, 2.1%, 1.3%, 1.3%, 2.1%, and 9.4% of the children already had early developmental delays in communication, gross motor, fine motor, problem-solving, personal-social, and social–emotional skills, respectively (Supplementary Table S1). Differences in the early development of each ECD domain were observed among the children with different demographic and family ecological characteristics (Table 4).

Table 4 ECD levels and differences among children with different demographic and family ecological characteristics (N = 534) Characteristic Communication GM skills FM skills PrS skills PeS skills SE development P -value P -value P -value P -value P -value P -value Total Sex 0.011 * 0.431 0.431 0.502 0.226 0.119 Preterm birth with oxygen supplementation at birth 0.462 0.219 0.456 0.037 * 0.265 0.994 Mother tends to ignore child while using digital devices 0.78 0.642 0.146 0.615 0.375 < 0.001 *** Mother use digital devices while caring for or conversing with child 0.501 0.178 0.1 0.016 * 0.007 ** 0.005 ** Number of sound-producing toys at home 0.604 0.319 0.519 0.819 0.255 0.021 * Number of adult books at home 0.041 * 0.166 0.435 0.352 0.017 * 0.052 Number of children’s books at home 0.201 0.001 ** 0.065 0.012 * 0.003 ** < 0.001 *** Mother’s frequency of reading newspapers or books 0.019 * 0.157 0.79 0.707 0.087 0.001** Child’s weekly frequency of screen exposure 0.018 * 0.175 0.552 0.408 0.159 0.98 Child’s average daily screen exposure time in the past week 0.344 0.046 * 0.905 0.551 0.64 0.569 Secure and safe level of family parenting environment 0.168 0.049 * 0.101 0.285 0.005 ** < 0.001 *** Frequency of parent-child shared activities 0.042 * 0.354 0.040 * 0.057 < 0.001 *** < 0.001 *** Frequency of marital arguments in the past 3 months 0.043 * 0.156 0.013 * 0.393 0.053 < 0.001 *** Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.012 *

0.411 0.003 ** 0.015 * 0.005 ** < 0.001 *** Family functioning (APGAR) score X 0.048 * X 0.722 X 0.005 ** X 0.183 X 0.036 * X <0.001 *** Maternal educational level 0.737 0.276 0.733 0.519 0.797 0.0497 *b Maternal employment status 0.329 0.444 0.582 0.072 0.233 0.014 * Household monthly income (CNY) 0.6 0.225 0.0499 *a 0.002 ** 0.182 0.012 * Initiated reading or storytelling to child within 6 months 0.041 * 0.677 0.052 0.097 0.524 0.527 Child's first exposure to screens within 6 months 0.61 0.484 0.945 0.676 0.031* 0.002 ** All demographic and family ecological factors were compared among different ECD status using Spearman test for one continuous variable and one ordinal variable, and Cochran-Mantel-Haenszel (CMH) statistic for binary, categorical, and ordinal variables in R×C tables. Complete details are provided in Supplementary Table S1 GM gross motor, FM fine motor, PrS problem-solving, PeS personal-social, SE social-emotional P -value The comparison was conducted by using CMH statistic test a The exact value was 0.0499 b The exact value was 0.0497 X P-value:The comparison was conducted by using Spearman test * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Family ecological influences on ECD

Boys and girls showed significant differences in early language development ($p = 0.001$). The probability of normal or better early language development in girls was more than twice that in boys (OR = 2.452). Early gross motor development was significantly different among children with different numbers of books ($p = 0.008$). For children with every additional 10 books, the early gross motor development level increased to 1.336 times that of those with ≤ 10 books (OR = 1.336).

For early fine motor development, no factor related to the family/parental environment was found to be statistically significant. Early problem-solving development was significantly related to whether the child was born prematurely and was accompanied by oxygen supplementation, the mother's marital happiness, and whether the mother used digital devices when caring for the child (all $p < 0.05$). Children who were born prematurely with oxygen supplementation were associated with a higher probability of abnormal early problem-solving development, however, the odds ratio required further validation, as the current number of children with this exposure was too small to calculate a stable estimate. Additionally, for every higher level of marital happiness, the probability of improving to normal early problem-solving development was twice that of early problem-solving development delay (OR = 2.091). Conversely, compared with early problem-solving developmental delay, those born prematurely or with oxygen supplementation had a 26 times higher probability of better early problem-solving development than those born prematurely with oxygen supplementation (OR = 26.591). For each higher level of marital happiness, the probability of better early problem-solving development was more than twice that of delayed early problem-solving development (OR = 2.315). However, the probability of a child with better problem-solving development whose mother used digital devices when caring for the baby was only 0.368 times that of those whose mothers did not use such devices when caring for the baby (OR = 0.368).

Early personal-social development was associated with the frequency of parent–child shared activities ($p = 0.046$) and whether mothers used digital devices when caring for their babies ($p = 0.026$). Children with an additional 1–2 times parent–child shared activities per week had 1.5 times normal or better personal-social development (OR = 1.524). If a mother used digital devices when caring for her baby, the probability of the child having normal or better personal-social development was only 0.78 times that of those whose mothers did not use such devices when caring for babies (OR = 0.780).

Early social–emotional development was significantly associated with frequency of parent–child shared activities (p < 0.001), whether mothers ignored their children when using digital devices (p = 0.042), frequency of spousal quarrels (p = 0.005), and family functioning (p = 0.035). The probability of children with additional 1–2 times parent–child shared activities per week was only 0.441 times that of early social–emotional developmental delay (OR = 0.441). For each one-point increase in the family functioning score, the odds of early social–emotional developmental delay were only 0.896 (OR = 0.896). However, for every higher level of neglect, a mother showed her child when using digital devices, and the probability of early social–emotional developmental delay of this child would increase by 1.214 times (OR = 1.214). Additionally, for every increase in the frequency of spousal quarrels, the probability of children experiencing early emotional-social developmental delays increased by 1.698 times (OR = 1.698) (Table 5).

Table 5 Logistic regression analyses of the influence of family parental environment on ECD Characteristic Coefficient Std. Error p -value OR 95% Wald CI Lower limit Upper limit Model 1: Early communication development Sex 0.897 0.28 0.001 ** 2.452 1.416 4.246 Number of adult books at home 0.048 0.107 0.652 1.049 0.851 1.293 Mother’s frequency of reading newspapers or books 0.292 0.188 0.119 1.34 0.928 1.935 Child’s weekly frequency of screen exposure -0.144 0.089 0.104 0.865 0.727 1.03 Frequency of parent-child shared activities 0.364 0.193 0.059 1.439 0.986 2.101 Frequency of marital arguments in the past 3 months -0.155 0.224 0.489 0.857 0.552 1.328 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.201 0.154 0.191 1.222 0.905 1.652 Family functioning (APGAR) score 0.005 0.061 0.937 0.995 0.882 1.122 Initiated reading or storytelling to child within 6 months -0.398 0.288 0.167 0.672 0.382 1.181 Model 2: Early gross motor development Number of children’s books at home 0.29 0.11 0.008 ** 1.336 1.077 1.658 Child’s average daily screen exposure time in the past week 0.367 0.192 0.056 1.443 0.991 2.101 Secure and safe level of family parenting environment 0.332 0.366 0.364 1.393 0.68 2.854 Model 3: Early fine motor development Frequency of parent-child shared activities 0.326 0.227 0.117 1.385 0.922 2.08 Frequency of marital arguments in the past 3 months -0.262 0.246 0.298 0.77 0.47 1.26 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.225 0.15 0.188 1.252 0.896 1.748 Family functioning (APGAR) score 0.029 0.41 0.676 1.03 0.897 1.182 Household monthly income (CNY) 0.188 0.129 0.145 1.207 0.937 1.553 Model 4: Early problem-solving development Model A: close to cut-off value vs. lower than cut-off value Preterm birth with oxygen supplementation at birth 10.287 1.184 < 0.001 *** 29361.344 2881.567 299173.556 Mother use digital devices while caring for or conversing with child -0.888 0.497 0.074 0.411 0.155 1.09 Number of children’s books at home -0.604 0.353 0.087 0.547 0.274 1.091 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.738 0.36 0.041 * 2.091 1.032 4.235 Household monthly income (CNY) 0.282 0.389 0.468 1.326 0.619 2.84 Model B: higher than cut-off value vs. lower than cut-off value Preterm birth with oxygen supplementation at birth 3.281 1.371 0.017 * 26.591 1.811 390.35 Mother use digital devices while caring for or conversing with child -1 0.473 0.035 * 0.368 0.145 0.93 Number of children’s books at home -0.102 0.296 0.729 0.903 0.506 1.611 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.839 0.315 0.008 ** 2.315 1.249 4.29 Household monthly income (CNY) 0.585 0.351 0.096 1.795 0.902 3.571 Model 5: Early personal-social development Mother use digital devices while caring for or conversing with child -0.249 0.112 0.026 * 0.78 0.627 0.97 Number of adult books at home 0.101 0.114 0.377 1.106 0.884 1.383 Number of children’s books at home 0.072 0.113 0.521 1.075 0.862 1.341 Secure and safe level of family parenting environment 0.328 0.389 0.399 1.388 0.648 2.976 Frequency of parent-child shared activities 0.421 0.211 0.046 * 1.524 1.008 2.303 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.161 0.152 0.291 1.174 0.871 1.583 Family functioning (APGAR) score 0.032 0.061 0.599 1.033 0.916 1.163 Child’s first exposure to screens within 6 months 0.229 0.313 0.464 1.258 0.681 2.324 Model 6: Early social-emotional

development Mother tends to ignore child while using digital devices 0.194 0.095 0.042 * 1.214 1.007 1.463 Mother use digital devices while caring for or conversing with child -0.014 0.101 0.893 0.986 0.809 1.203 Number of sound-producing toys at home 0.083 0.12 0.486 1.087 0.859 1.375 Number of children’s books at home -0.072 0.087 0.408 0.931 0.785 1.104 Mother’s frequency of reading newspapers or books -0.19 0.138 0.168 0.827 0.631 1.084 Secure and safe level of family parenting environment -0.2 0.319 0.53 0.819 0.438 1.53 Frequency of parent-child shared activities -0.819 0.184 < 0.001 *** 0.441 0.308 0.633 Frequency of marital arguments in the past 3 months 0.529 0.189 0.005 ** 1.698 1.172 2.459 Self-rated marital happiness [Scale: 1 (lowest) to 7 (highest)] 0.013 0.127 0.917 1.013 0.789 1.301 Family functioning (APGAR) score -0.11 0.052 0.035 * 0.896 0.809 0.992 Maternal educational level -0.004 0.153 0.981 0.996 0.738 1.345 Maternal employment status 0.043 0.273 0.875 1.044 0.612 1.781 Household monthly income (CNY) -0.093 0.106 0.38 0.911 0.74 1.122 Child’s first exposure to screens within 6 months -0.447 0.256 0.081 0.64 0.387 1.057 Model 1 was the ordinal logistic regression model, which was supported by the parallelism test (χ 2 = 17.172, P = 0.050). The likelihood ratio test showed χ 2 = 32.573 (P < 0.001), indicating the model fit the data well. Diagnostic tests suggested that no significant multicollinearity was detected, as all VIF values were below the threshold of 1.7. Model 2 was the ordinal logistic regression model, which was supported by the parallelism test (χ 2 = 2.882, P = 0.410). The likelihood ratio test showed χ 2 = 17.265 (P = 0.001), indicating the model fit the data well. Diagnostic tests suggested that no significant multicollinearity was detected, as all VIF values were below the threshold of 1.2. Model 3 was the ordinal logistic regression model, which was supported by the parallelism test (χ 2 = 4.131, P = 0.531). The likelihood ratio test showed χ 2 = 15.676 (P = 0.008), indicating the model fit the data well. Diagnostic tests suggested that no significant multicollinearity was detected, as all VIF values were below the threshold of 1.6. Model 4 was the multinomial logistic regression model rather than ordinal logistic regression model, because the model did not pass the parallelism test (χ 2 = 20.287, P = 0.001). The likelihood ratio test showed χ 2 = 34.228 (P < 0.001), indicating the multinomial logistic regression model fit the data well. Diagnostic tests suggested that no significant multicollinearity was detected, as all VIF values were below the threshold of 1.1. Model 5 was the ordinal logistic regression model, which was supported by the parallelism test (χ 2 = 2.105, P = 0.978). The likelihood ratio test showed χ 2 = 30.065 (P < 0.001), indicating the model fit the data well. Diagnostic tests suggested that no significant multicollinearity was detected, as all VIF values were below the threshold of 1.6. Model 6 was the ordinal logistic regression model, which was supported by the parallelism test (χ 2 = 19.746, P = 0.138). The likelihood ratio test showed χ 2 = 99.758 (P < 0.001), indicating the model fit the data well. Diagnostic tests suggested that no significant multicollinearity was detected, as all VIF values were below the threshold of 1 * p < 0.05, ** p < 0.01, *** p < 0.001.

Discussion

Our findings showed that 16.7% of children had confirmed developmental delay in at least one developmental domain. Boys born prematurely with oxygen supplementation, children without enough child books, parents with a lower level of marital happiness, fewer weekly parent–child shared activities, mothers’ screen use behaviors, neglect when using screens, and family functioning were associated with developmental delays in young children. Preterm birth with supplemental oxygen, mothers’ responsive caregiving behaviors in the digital era, and family functioning emerged as novel and critical factors correlated with ECD. We also provide empirical evidence to address the current scarcity of comprehensive evidence on multiple risk factors linked to all domains of ECD [17], especially for young children in urban regions.

Unlike previous studies conducted in rural settings, we focused on ECD in a metropolitan area. Compared to the extremely scarce data on the ECD delay rate among 1–36-month children in urban areas of China [83], the prevalence of early developmental delays in at least one domain was relatively higher. Focusing on the delay rate in a single domain, the language developmental delay rate here was lower than that of children under 6 years in Shanghai [84] and 0–18-month children in Guangzhou [85]; however, 8.2% of children in this study were evaluated as having suspected developmental delay. The delay rates of gross motor, fine motor, and personal-social skills were lower than those of children in Guangzhou [85]. However, these data were derived from different child populations across various studies, and there is a lack of research that comprehensively reports both overall ECD delay rate and the respective delay rates in each specific domain.

Although such data for large cities in China are limited, the prevalence among children in rural areas has been extensively reported. A systematic review concluded that there was an approximately 15% delay rate of motor development, 26–72% language developmental delay, 22–57% cognitive delay, and 14–59% social–emotional delay in rural China [86]. Furthermore, although one study reported data on social–emotional developmental delay in 0–3-year-old children, it focused on a very specific population [87] while another study targeted 0–6-month infants in rural China [42]. In urban areas, only Hangzhou reported a social–emotional developmental delay rate for children aged 3–5 years [88]. There is a paucity of data on the prevalence of developmental delays in problem-solving skills among young children.

To further understand ECD delay in 0–3-year-old children from Shanghai, the associated factors should be explored. Previous studies focused on a single factor [89 – 91] or factors in a single subsystem [49 , 92 – 96]; however, the consideration of factors correlated with ECD should be placed within a broader context beyond the immediate microsystem [21]. For example, parental investment and educational level are jointly linked to ECD, which aligns with the core concept of Bronfenbrenner’s family ecosystem [22 , 23]. Furthermore, an increasing number of international studies have emphasized that family centered parental caregiving plays a core role in ECD promotion [21 , 97 – 101]. The United States has proposed a new family centered and multilevel conceptual model [101]. The key factors correlated with ECD identified here all fell within the microsystem of the family ecosystem model, whereas factors from the exosystem did not emerge as significant predictors in the specific context of Shanghai, China’s most economically developed metropolis. This distinction sets the present findings apart from prior studies conducted in rural China, where factors beyond the microsystem, such as parental and family socioeconomic characteristics [39 , 42 , 46 , 102], and intergenerational support [14], were found to be significantly associated with ECD outcomes.

The negative effect of the male gender on early communication development was consistent with evidence reported from diverse regions, both domestically and internationally. For example, some studies have reported that boys consistently demonstrate poorer outcomes in multiple ECD domains, including communication skills [6 , 38 , 95 , 103], while others have reported a specifically elevated risk of language development delays in boys [51 , 104].

Considering preterm birth, many studies have consistently highlighted significant adverse effects on early development of language [32 , 105 , 106], cognition [105 – 107], motor [105 – 108], and overall development [109]. While we did not find such an association, we explored for the first time the impact of preterm birth with oxygen supplementation on ECD. This characteristic was significantly associated with poorer early development of problem-solving skills. There is limited evidence on the impact of oxygen supplementation at birth on the early development of problem-solving skills.

However, a 2024 Cochrane study protocol raised the core question of whether supplemental oxygen could reduce long-term neurodevelopmental impairments in late preterm infants [110]. This intervention protocol is the first to highlight the importance of oxygen supplementation at birth for ECD delays. Although this publication only presents the study protocol and lacks final evidence and specific data, its primary objective supports the findings of this study, corroborating the rationale of the research protocol and its core aim.

The importance of the number of children's books was also observed in this study. Although the association between the number of children's books and the development of language [111] and social–emotional skills [48] has been demonstrated in other studies, this study did not find such a relationship, which may be attributed to rural differences [48 , 103] in China or racial differences among countries [111]. In contrast to previous findings, our results suggest an association between the number of children's books and the early development of gross motor skills. This finding is consistent with prior work reporting that book availability is linked to overall ECD outcomes, including gross motor development [10 , 38 , 112]. Notably, the number of children's books may not directly promote gross motor development but rather serves as a proxy for a broader nurturing home environment. Book-rich households often provide diverse play materials and outdoor opportunities, and shared reading activities can incorporate physical actions, indirectly supporting gross motor skills [113]. Therefore, this finding should be interpreted as a reflection of overall family interaction patterns and stimulating environments, rather than a direct causal effect of book quantity on motor development [114 , 115]. Research in rural China has shown that books alone do not benefit child development when caregivers lack early reading knowledge and a daily interactive routine [10]. Interventions should shift from book access to high-quality daily reading practices—for example, through group training and a daily routine of at least 15 min of dialogic book sharing [34 , 116].

The nuclear family centers on two key relationships: the parent–child relationship and the marital relationship, which are closely linked to ECD. The positive effect of frequent parent–child shared activities on the early development of personal-social and social–emotional skills identified in this study was consistent with many other theories and studies. The effect of parental investment, especially parent–child interaction time, is more positive than that of financial investment [117]. Parent–child shared activities, including shared reading or book sharing, storytelling, playing, singing, and drawing, are directly linked to ECD [118 – 121]. Studies in different countries have pointed out the positive effect of a greater frequency of parent–child shared activities on overall ECD outcomes [34 , 37 , 45] and the early development of communication [36 , 58 , 122 , 123], cognitive [34 , 35 , 58], motor [34 , 36 , 58], personal-social [36], and social–emotional [34 , 58 , 124 , 125] skills, which supports the findings of this study. Among the various parent–child shared activities, particular emphasis has been placed on shared reading or book sharing, which could provide an extra and significant basis for prosocial behaviors and good peer relationships [126 – 129], and bring more joy [130], benefiting a child's whole life. Therefore, given the flexibility of conducting parent–child shared activities in various settings (anytime and anywhere) [131], increasing their frequency merits prioritization in every household [132].

However, low levels of marital happiness and more frequent marital arguments were associated with higher risks of early developmental delays in problem-solving and social–emotional skills. These two marital relationship characteristics have not been extensively examined in nuclear families. Regarding the negative impact of marital arguments on the early development of socio-emotional skills, studies from both paternal and maternal perspectives in American families have yielded relatively consistent conclusions [12 , 133], which align with the findings of this study. Concerning marital happiness, a survey of 89 families in the United States found that the children of parents with

a higher level of marital happiness exhibited better development of socio-emotional skills [134], which has also been validated in a study of 212 families with 1–3-year-old children in Shanghai, China [135]. However, current research on marital relationships primarily focuses on their impact on early childhood socio-emotional development, while studies on other domains are notably scarce. In early parenthood, first-time parents experience a decline in marital happiness [136], and this change is correlated with poorer early problem-solving skills [137], which also supports the findings of this study. Nevertheless, there is insufficient evidence regarding the impact of marital arguments on socio-emotional development and the influence of marital happiness on problem-solving skills within a localized context.

Furthermore, it is important to recognize that conflict is inevitable in these family relationships. However, as pointed out by the McMaster family functioning model, how a family thinks about and handles conflicts when they arise is more critical [29], as it reflects family quality and demonstrates mutual support and adaptability among family members [64]. This high level of family support and adaptability was correlated with reduced odds of early social–emotional developmental delays. However, research on the relationship between family functioning and ECD delays is scarce. Only one study in rural China mentioned an association between parenting quality and developmental delay; however, it did not assess family quality using a standard family functioning scale [39]. Therefore, the relationship between family functioning and ECD, particularly in the urban context, requires further exploration and validation.

Responsive caregiving highlighted by the Nurturing Care Framework is extremely significant for ECD [3]. A community-based responsive caregiving program significantly improved the development of cognition, fine motor, gross motor, and language [138], and a survey in Beijing demonstrated that the absence of responsive caregiving predicted higher odds of overall ECD delay among children aged 24–59 months [139]. However, responsive caregiving behaviors may be influenced by digital era features, as caregivers and their children are increasingly exposed to the use of digital devices [140], which may influence ECD outcomes. Consequently, research has focused on the impact of screen exposure on ECD and reached relatively consistent conclusions [141 , 142], demonstrating that excessive daily screen exposure (> 1 h) is significantly associated with poorer cognition, language, and social–emotional skills [58 , 143 – 146]. This aligns with the WHO’s recommended limit for children’s screen exposure of < 1 h per day [146]. However, screen exposure was not a statistically significant risk factor for 0–36-month-old children in Shanghai, which may be attributed to the fact that only 5.3% of the participating children were exposed to screens for > 1 h per day. Therefore, investigating only the effects of screen exposure on ECD in young children is of limited research value.

In contrast to children, caregivers have inevitably increased use of digital devices [140]. However, whether this increase would lead to changes in responsive caregiving behaviors and whether such changes impact ECD has not received previous attention, with only one systematic review mentioning potential adverse psychosocial effects [147]. We, therefore, revealed for the first time that while parenting, nearly 60% of mothers used digital devices, and approximately 30% completely or almost completely neglected their children during use. This alteration in responsive caregiving due to digital device use was associated with a significantly elevated risk of developmental delays. Such long-term negative consequences of neglect on children’s health are consistent with previous research [9 , 96 , 148].

Although the regression models did not show statistical significance, the correlation analyses revealed other associations: between household monthly income, maternal educational level, and

maternal employment status from the exosystem; factors of initiated reading or storytelling to children within 6 months; children's first exposure to screens within 6 months from the microsystem; and early development of communication, fine motor, problem-solving, personal-social, and socio-emotional skills. These factors have been identified in low- and middle-income countries [13 , 149 – 152]; however, whether these factors could exert a significant effect in Shanghai requires further investigation.

Regarding intergenerational support, no positive results were found. This is consistent with the contradictory conclusions reported across different regions and cultural contexts. Studies in rural China found that grandparental support was associated with poorer cognitive, language, and socio-emotional development [14]; research in the United States linked grandparental support to better language development [52]; and studies in rural Pakistan associated grandparental support with better cognitive, fine motor, and socio-emotional development [53]. Therefore, the impact of this macrosystem factor on ECD in urban China, particularly in its most economically developed regions, warrants further exploration.

This study has several limitations. First, due to the use of convenience sampling, all participants were recruited from Baoshan District, Shanghai, which limits the generalizability of the findings to other regions. Compared with the general population of Baoshan District, our sample overrepresented higher-income and higher-educated families. Additionally, the child's age distribution of our sample differed from that of the source population, indicating age-related selection bias. Therefore, generalization of our findings to families with lower socioeconomic status or to younger infants and older toddlers should be made with caution. Second, as a cross-sectional study, we can only identify factors associated with ECD outcomes and cannot infer causality. Moreover, constrained by the study design, we were unable to assess the influence of the chronosystem (e.g., dynamic changes in parenting practices over time) on ECD. Third, limited by data accessibility and pilot survey feedback, this study did not include some factors from the mesosystem or exosystem (e.g., access to childcare institutions and community services), nor did it include cultural or value-related factors from the macrosystem, because participants generally reported that they paid little attention to these aspects and found them difficult to answer accurately. Furthermore, we only analyzed the independent associations of various factors within the family ecosystem with ECD, without examining interactions among different subsystems or modeling causal pathways among factors, mainly due to the cross-sectional design and sample size constraints. Future studies could employ structural equation modeling or directed acyclic graphs to further explore the complex cross-subsystem mechanisms within the family ecosystem and their pathways of influence on ECD.

Children aged 1–36 months in Shanghai face a relatively severe risk of ECD delays. Fortunately, the factors correlated with these delays demonstrate good observability and high potential for intervention. This study is the first to identify the significant roles of preterm birth with supplemental oxygen, mothers' responsive caregiving behaviors in the digital era, and family functioning. Early attention to language development milestones, functioning, and timely interventions in boys, along with increased investment in children's books, are crucial to effectively improve early language and fine motor development outcomes. For children born preterm on supplemental oxygen, it is essential to increase the frequency of developmental monitoring, specifically in the problem-solving domain, and enhance primary caregiver knowledge of relevant milestones and effective stimulation approaches. Furthermore, child healthcare providers should prioritize follow-up and interventions for children from families with low marital happiness for early development of problem-solving skills; with low parent–child interaction for early development of personal-social skills; and characterized by low parent–child interaction, high marital conflict, and low family quality for early development of

social–emotional skills. Finally, addressing the diminution of responsive caregiving due to caregivers' engagement with contemporary digital technologies is a critical issue demanding ongoing vigilance from individual families, pediatric healthcare providers, and broader societal and national policies.

Supplementary Information

Supplementary Material 1